Task – 8

Cricket Analysis Report – Data Ingestion Using URL

T20 International Team Performance Analysis – 03 March 2026

Tool Used: Power BI

Task Description

Objective:
To ingest cricket statistics data from a public URL into Power BI and create interactive visualizations for performance analysis.

Scenario:
Given a Cricinfo Statsguru URL, the task was to:

- Load data into Power BI using Web connector
- Extract structured HTML table data
- Clean and transform the dataset
- Create calculated measures using DAX
- Build an interactive dashboard using different chart types

Tools Used:

- Power BI Desktop
- Web Data Connector
- Power Query (M Language)
- DAX (Data Analysis Expressions)

Dataset Overview:

- The dataset was extracted from ESPN Cricinfo (T20 International records).
- The dataset includes:
- Team
- Runs Scored
- Wickets Lost
- Balls Faced
- Batting Average
- Run Rate
- Match Result (Won/Lost/Tied)
- Opposition
- Ground

- Start Date

Total records analysed: **50 matches**

The data was dynamically extracted from HTML tables and cleaned using Power Query.

Data Cleaning & Transformation

The following steps were performed:

- Extracted table using Html.Table() in Power Query
- Renamed columns for clarity
- Removed null and unwanted rows
- Converted data types (numeric, date, text)
- Created DAX measures:
    - Total Matches
    - Total Wins
    - Total Losses
    - Win Percentage

This ensured the dataset was structured and ready for visualization.

Dashboard Visualizations & Explanation

1. **KPI Cards**

- **Total Matches:** Displays total matches played.
- **Total Wins:** Shows total number of victories.
- **Total Losses:** Displays total losses.
- **Win Percentage:** Shows overall success rate.

  These KPIs provide quick performance summary.

2. **Match Outcome Distribution (Donut Chart)**

   Displays distribution of:

- Wins
- Losses
- Tied matches

  This helps understand overall performance consistency.

3. **Run Rate vs Runs (Scatter Chart)**

   Shows relationship between:

- Total Runs
- Run Rate

  Observation:

  Higher run totals generally correspond to higher run rates, indicating aggressive scoring strategies.

4. **Top Performances by Opposition (Bar Chart)**

   Displays maximum runs scored against different opponents.

   This helps identify strongest batting performances against specific teams.

5. **Runs Distribution**

   Shows frequency of different score ranges.

   This indicates scoring consistency and high-scoring match patterns.

Key Insights & Observations

1. **Strong Overall Performance (84% Win Rate)**

   The team has achieved a win percentage of **84%** across 50 matches. This indicates a highly consistent and dominant performance in T20 Internationals.

   Such a high success rate reflects:

   - Effective batting strategies
   - Strong match execution
   - Ability to convert good scores into victories

   The low number of losses suggests strong control in most match situations.

2. **High Scores Strongly Correlate with Wins**

   The analysis shows that most matches with higher total runs resulted in wins.

   This indicates:

   - Batting performance plays a critical role in match outcomes.
   - When the team scores above 200 runs, the probability of winning significantly increases.
   - Aggressive scoring strategies lead to better match control.

   This suggests that maintaining high scoring momentum is a key success factor.

3. **Run Rate as a Performance Indicator**

   The scatter analysis between **Runs and Run Rate** shows a positive relationship:

   - Higher runs are typically associated with higher run rates.
   - Efficient scoring per over increases match-winning probability.

   This confirms that not just total runs, but scoring efficiency (Run Rate) impacts match results.

4. **Opposition-Based Performance Strength**

   The bar chart showing maximum runs against opposition indicates:

- The team has recorded strong performances against multiple opponents.
- Certain teams appear more vulnerable to high scoring totals.

- Performance consistency across different opponents reflects tactical adaptability.

  This highlights strategic flexibility against varied bowling attacks.

5. **Consistency Across Venues**

   By analyzing matches by ground:

- Strong performance is observed across multiple venues.
- The team is not dependent on a single home ground advantage.
- Adaptability to different pitch conditions is evident.

  This strengthens overall team credibility in international tournaments.

6. **Limited Tied Matches (Decisive Outcomes)**

   The dataset shows very few tied matches, indicating:

- Matches are mostly decisively won or lost.
- The team often maintains control during critical match phases.
- Strong finishing ability in close situations.

7. **Balanced Risk-Reward Strategy**

   From scoring distribution:

- Majority of matches fall within a stable scoring range.
- Few extreme outliers suggest controlled aggressive batting rather than reckless play.
- This balance contributes to consistent success.

Conclusion

This project successfully demonstrates the process of ingesting real-world cricket performance data from a web-based source into Power BI and transforming it into meaningful analytical insights. The task involved extracting structured data from an HTML table, performing data cleaning and transformation using Power Query, and developing calculated measures using DAX to support performance evaluation.

The analysis reveals that the team has maintained a strong competitive edge in T20 International matches, reflected by an impressive **84%-win rate**. The data indicates that higher total scores and improved run rates significantly increase the probability of winning matches. This suggests that aggressive yet controlled batting strategies play a crucial role in overall match success.

Furthermore, the dashboard highlights consistent performance across different opponents and venues, demonstrating adaptability and strategic strength. The relationship between scoring efficiency and match outcomes reinforces the importance of maintaining momentum throughout the innings.

From a technical perspective, this project showcases:

- Effective use of Web data ingestion in Power BI
- HTML table extraction using M language
- Data transformation and cleaning best practices
- Creation of DAX measures for KPI analysis
- Development of an interactive and professional dashboard

Overall, the dashboard provides a dynamic and interactive analytical framework that enables quick decision-making and performance monitoring. The visualizations offer clear insights into scoring patterns, win distribution, and efficiency metrics, making the analysis both informative and actionable.

This project successfully combines technical implementation with analytical interpretation, resulting in a comprehensive performance evaluation model using Power BI.

Recommendations

- Continue aggressive batting strategy in high-scoring conditions.
- Analyze opposition-specific strategies for continuous improvement.
- Monitor run rate trends for performance optimization.
- Use venue-based analysis for strategic planning.

Final Summary

This task successfully demonstrated:

- Web data ingestion in Power BI
- HTML table extraction using M language
- Data cleaning and transformation
- DAX measure creation
- Professional dashboard development

The final output provides meaningful insights through interactive visualizations.

- The team demonstrates strong dominance in T20 Internationals.
- High scoring and efficient run rate are key success drivers.
- Performance consistency across opponents and venues enhances competitive strength.
- Strategic aggression and match control contribute significantly to the high win percentage.